Lecture 3

Tuesday, 17 January 2023 1:25 PM

HW1 (due Thursday)

HW0, part 1 → Course evaluation (due Friday)

↳ Heterogenous reaction (e^- transfer between electrodes)

→ written as half-reactions (oxidation/reduction)

→ short hand notation: e^- chemical cell

ex: Chlor-Alkali

$2Cl^-_{(aq)} \rightarrow Cl_2(g) + 2e^-$

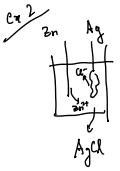
$2H_2O + 2e^- \rightarrow 2OH^- + H_2(g)$



$Pt \mid Cl_2(g) \mid Cl^-, Na^+ \,\|\, Na^+, OH^-, H_2O \mid H_2(g) \mid Pt$

↑ denotes change in composition (although in the same phase) (separator)

ex 2



$Zn(s) \mid Zn^{2+}, Cl^- \mid AgCl(s) \mid Ag$

Electrochemical Concepts and Conventions

+ At least 2 electrodes

↳ Anode → Oxidation reaction (An-Ox)

$R \rightarrow O + e^-$

↑ reductant (gets oxidized)

↳ Cathode → Reduction reaction

$O + e^- \rightarrow R$

Any phase in which ions can be transported.

+ Electrodes must be connected via an electrolyte

+ Transition between electric/ionic conduction occurs @ electrode/electrolyte interface

+ Electroneutrality states that currents @ electrodes must be equal and opposite

$$I_{anode} + I_{cathode} = 0$$

Convention: $I_{anode} \geq 0$

+ Cell potentials (Galvanic + Electrolytic Cell)

↳ Galvanic cell: half-reactions sum to an overall spontaneous rxn ($\Delta G_r < 0$)

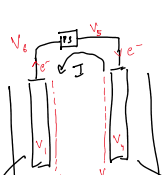
+ We can extract work from these types of cells (batteries)

+ H_2-fuel cell: $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$

↳ Electrolytic cell: electrical power from an external source is required to make rxns proceed ($\Delta G_r > 0$)

+ Put work into the cell

+ Charging a battery, water electrolysis.



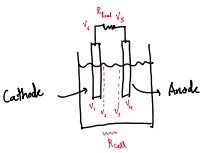
+ Requires power supply (PS)

+ Current is reversed

+ Now RHS electrode is cathode

+ PS increases $(V_6 - V_5) > V_{oc}$

Current flows from $V_6 \rightarrow V_5$ (high to low potential)



+ Anode and cathode are at different potentials since diff rxns occur @ them

+ I_{anode} → positive (convention)

∴ current flows opposite of electrons

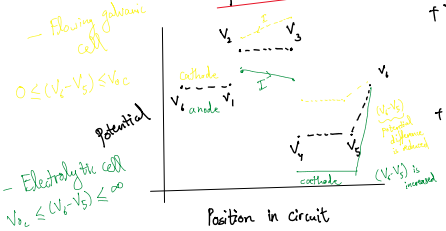
+ Ohm's law: $V = IR$

$(V_6 - V_5) = I R_{load}$

↳ If $R_{load} \gg R_{cell}$ → No current in external circuit

↳ $(V_6 - V_5)$ then describes/defines open circuit potential (VOC)

Measured by connecting e^- w/ a large known resistor & measuring small current



+ If conductive wires:

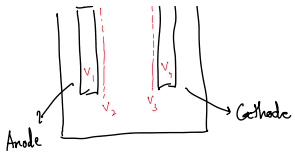
1) $V_6 \approx V_1$

2) $V_5 \approx V_4$

+ @ open circuit, $V_2 \approx V_3$

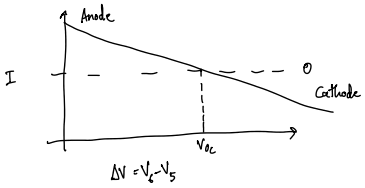
↳ No resistance to flow [$(V_6 - V_5) \approx 0$]

Max current but no work extracted → short circuit/closed circuit



+ Now RHS electrode is cathode
+ PS increases $(V_6 - V_5) \geq V_{oc}$
  Current flows from $V_6 \rightarrow V_5$ (high to low)
  [Opp. to potential]
  ↓
  open-circuit
  (at open-circuit there is no net current flow $I=0$)

→ Cathode & Anode switching is confusing



So, even though $I_{anode} > 0$, there can be both oxidation & reduction reactions that can occur, but just the net should be oxidation.

↳ No resistance to flow [V_6 "5"]
  Max current but no work extracted → short circuit / closed circuit

↳ Therefore, to achieve max power ($P = I(V_6 - V_5)$) we need an intermediate load

